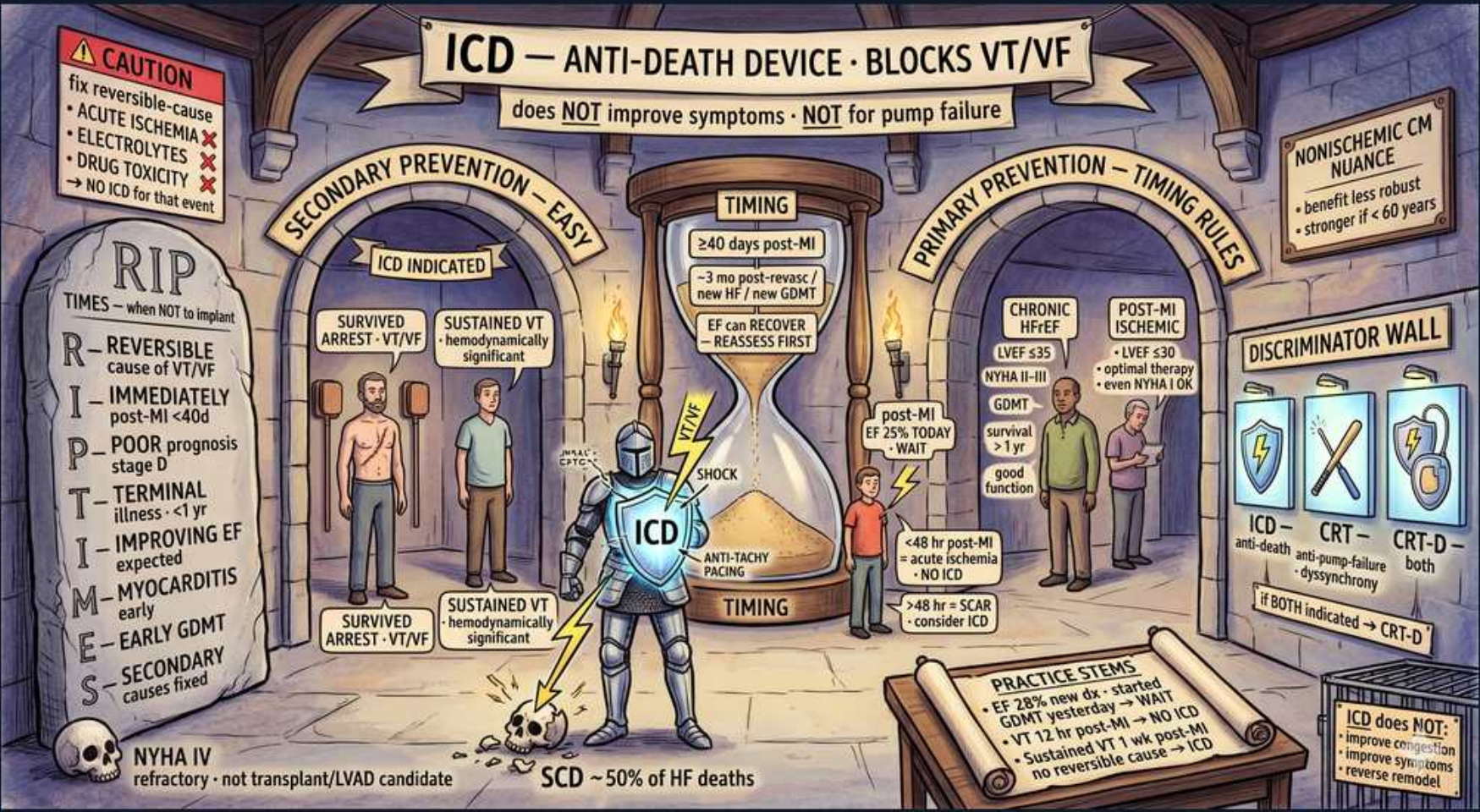


Heart Failure — ICD (Anti-Death Device)



1. ICD blocks VT/VF, not pump failure

WHAT YOU SEE

Title: ICD anti-death, blocks VT/VF, NOT symptoms/pump

WHAT IT ENCODES

The ICD is purely an 'anti-death' device: it senses sustained VT/VF and delivers anti-tachycardia pacing or a defibrillating shock to abort sudden cardiac death. It does NOT improve symptoms, exercise capacity, ejection fraction, or reverse remodeling — that is the job of GDMT and CRT. Mechanistically it converts a potentially-arrhythmic death into a non-sudden (pump-failure) death down the line, so it only helps patients whose competing risk of dying from pump failure is not already dominant. Mnemonic: ICD = 'shock box,' not a 'feel-better box.'

BOARD TRAP

WRONG answer: 'an ICD will improve this patient's dyspnea and functional class.' Discriminator: ICDs reduce arrhythmic mortality only; symptom/EF improvement comes from GDMT (ARNI, β -blocker, MRA, SGLT2i) and CRT. If the stem asks what improves symptoms in HFrEF, never pick the ICD.

2. Secondary prevention — easy

WHAT YOU SEE

Secondary prevention column

WHAT IT ENCODES

Secondary prevention is the slam-dunk indication: a patient who has ALREADY survived a cardiac arrest from VT/VF, or who has hemodynamically significant sustained VT (with or without structural heart disease), gets an ICD regardless of EF. No GDMT waiting period and no EF threshold is required because the event itself defines the risk. This is the one setting where EF can be normal and an ICD is still indicated.

BOARD TRAP

WRONG answer: 'wait 3 months on GDMT and reassess EF before implanting' in a survived-VF patient. Discriminator: the GDMT-optimization and EF-threshold rules apply only to PRIMARY prevention; a documented prior VT/VF arrest mandates a secondary-prevention ICD now.

3. Rule out reversible cause

WHAT YOU SEE

RIP / reversible-cause caution wall

WHAT IT ENCODES

Before any ICD, exclude a fully REVERSIBLE trigger of the arrhythmia: acute ischemia/MI within 48 hours, electrolyte derangement (hypokalemia, hypomagnesemia), drug toxicity (QT-prolonging agents, digoxin), or a transient metabolic insult. If the cause is reversible and correctable, fix it instead — an ICD is not indicated because the arrhythmia is not expected to recur once the trigger is gone. Mnemonic on the wall: the 'RIP' caution list = Reversible, Immediately post-MI, POOR prognosis, Terminal, Improving EF, Myocarditis, early GDMT, Secondary-fixed.

BOARD TRAP

WRONG answer: 'implant an ICD' for VF occurring during an acute STEMI. Discriminator: VT/VF within the first 48 h of an MI is from acute ischemia (a reversible cause) and does NOT by itself qualify for an ICD; revascularize and reassess later.

4. Primary prevention — timing rules

WHAT YOU SEE

Primary prevention column

WHAT IT ENCODES

Primary prevention targets patients who have NOT yet had a sustained arrhythmia but are at high risk because of severe LV dysfunction. The core threshold is LVEF $\leq 35\%$ on optimal GDMT (NYHA class II–III), or LVEF $\leq 30\%$ with NYHA class I; CRT-D is used when CRT criteria are also met. The timing rules (waiting periods by etiology) exist because EF often recovers with GDMT or revascularization, which would make the device unnecessary.

BOARD TRAP

WRONG answer: 'implant primary-prevention ICD based on a single low EF at presentation.' Discriminator: primary-prevention indication requires EF reassessment AFTER an adequate course of GDMT (≥ 3 months) and after the etiology-specific MI/revascularization waiting period — not the index EF.

5. >40 days post-MI

WHAT YOU SEE

Hourglass: ≥ 40 days post-MI

WHAT IT ENCODES

For ischemic cardiomyopathy you must wait ≥ 40 days after an MI (and ≥ 90 days after revascularization, CABG or PCI) before a primary-prevention ICD. The DINAMIT and IRIS trials showed NO survival benefit — and a shift toward non-arrhythmic death — when ICDs were implanted in the early post-MI window. The hourglass marks this mandatory delay.

BOARD TRAP

WRONG answer: 'implant an ICD now to protect this low-EF patient through the high-risk early post-MI period.' Discriminator: early post-MI ICD (DINAMIT/IRIS) gives no mortality benefit; the correct answer is to optimize GDMT, treat ischemia, and reassess EF at ≥ 40 days (or ≥ 90 days post-revascularization).

6. Reassess EF first

WHAT YOU SEE

EF can recover — reassess first

WHAT IT ENCODES

The whole point of the waiting period is that EF frequently RECOVERS on GDMT and after revascularization — a stunned/hibernating myocardium can regain function. Reassess LVEF with repeat echo at the end of the waiting window; if EF has climbed above the threshold ($>35\%$), the ICD is no longer indicated. Implanting before reassessment exposes the patient to device risks (infection, lead complications, inappropriate shocks) with no benefit.

BOARD TRAP

WRONG answer: 'the EF at the time of MI is what determines ICD eligibility.' Discriminator: eligibility is based on the EF measured on optimal medical therapy after the waiting period, not the acute/index EF, precisely because recovery is common.

7. Chronic ischemic HFrEF

WHAT YOU SEE

Chronic ischemic, LVEF ≤ 30 / ≤ 35 NYHA I-III

WHAT IT ENCODES

For chronic (stable, remote-MI) ischemic cardiomyopathy the evidence is strongest: MADIT-II supports an ICD at LVEF $\leq 30\%$ (even NYHA I), and SCD-HeFT supports LVEF $\leq 35\%$ with NYHA class II–III. These patients have a fixed scar substrate that sustains re-entrant VT, so the device provides a clear, durable mortality benefit — unlike the early post-MI or NICM settings.

BOARD TRAP

WRONG answer: 'NYHA IV chronic ischemic patient should get a standard primary-prevention ICD.' Discriminator: the chronic-ischemic indication covers NYHA I–III (per MADIT-II/SCD-HeFT); ambulatory NYHA IV is generally excluded unless the patient is a CRT-D, transplant, or LVAD candidate.

8. Nonischemic CM nuance

WHAT YOU SEE

Nonischemic CM nuance box

WHAT IT ENCODES

In nonischemic dilated cardiomyopathy (NICM) the benefit is LESS robust. The DANISH trial found no significant all-cause mortality reduction from primary-prevention ICDs in NICM, though a prespecified subgroup of younger patients (roughly <60 – 68 years) did benefit, likely because they have fewer competing non-sudden causes of death. Require ≥ 3 months of optimal GDMT and individualize the decision; many NICM patients on modern GDMT recover EF.

BOARD TRAP

WRONG answer: 'ICD primary prevention benefits NICM patients as strongly as ischemic patients.' Discriminator: DANISH showed the mortality benefit in NICM is uncertain overall and concentrated in younger patients; don't equate NICM with ischemic CM for ICD benefit.

9. >3 mo on GDMT (NICM)

WHAT YOU SEE

3 month GDMT post-onset NYMA/EF reassess

WHAT IT ENCODES

For nonischemic (and newly-diagnosed) cardiomyopathy, give ≥ 3 months of optimal four-pillar GDMT (ARNI/ACEi, evidence-based β -blocker, MRA, SGLT2i) before reassessing EF and deciding on a primary-prevention ICD. Many patients reclassify out of the $\leq 35\%$ threshold once therapy is titrated. Premature implantation is the classic exam error — 'don't implant too early.'

BOARD TRAP

WRONG answer: 'newly diagnosed NICM with EF 25% — implant an ICD today.' Discriminator: a freshly diagnosed cardiomyopathy needs ≥ 3 months of optimized GDMT and EF reassessment first; the correct next step is to titrate medical therapy, not implant immediately.

10. ICD does NOT improve symptoms

WHAT YOU SEE

Bottom-right: ICD does not improve symptoms/QOL/remodeling

WHAT IT ENCODES

Reinforcing the central concept: an ICD does not improve symptoms, quality of life, or remodeling, and can even worsen QOL through inappropriate shocks and device-related anxiety. Its only proven benefit is reducing arrhythmic (sudden) death. Pair it with CRT ($\rightarrow$ CRT-D) when symptomatic improvement and resynchronization are also desired.

BOARD TRAP

WRONG answer: 'add an ICD to improve quality of life in a symptomatic HFrEF patient.' Discriminator: QOL and symptom benefit come from GDMT and CRT; an ICD is neutral-to-negative for QOL and is chosen only for sudden-death prevention.

11. NYHA IV refractory — no ICD

WHAT YOU SEE

Bottom-left: NYHA IV refractory, not transplant/LVAD candidate

WHAT IT ENCODES

Avoid a stand-alone ICD in refractory, ambulatory NYHA IV / Stage D heart failure unless the patient is a candidate for transplant, LVAD, or CRT-D. In end-stage disease the competing risk of pump-failure death dominates, so preventing arrhythmic death merely converts the mode of dying without extending meaningful survival. Expected survival <1 year (non-bridge) is a contraindication.

BOARD TRAP

WRONG answer: 'every end-stage / NYHA IV heart failure patient should receive an ICD.' Discriminator: in refractory NYHA IV without a transplant/LVAD/CRT plan, an ICD offers no survival benefit because non-sudden death predominates — focus on advanced therapies or palliative care instead.

12. SCD ~50% of HF deaths

WHAT YOU SEE

SCD ~50% of HF deaths skull

WHAT IT ENCODES

Sudden cardiac death — usually from VT/VF — accounts for roughly half of all heart-failure deaths, and SCD is proportionally MORE common in milder HF (NYHA II) while pump-failure death dominates in advanced HF (NYHA IV). This inverse relationship is the rationale for offering ICDs earlier in the disease and explains why advanced-HF patients derive less ICD benefit.

BOARD TRAP

WRONG answer: 'sudden cardiac death is most likely in the sickest NYHA IV patients, so they benefit most from ICDs.' Discriminator: the PROPORTION of deaths that are sudden is highest in less-symptomatic patients; NYHA IV patients die mostly of pump failure, so they benefit least from an ICD.

13. CRT-D if both indicated

WHAT YOU SEE

Discriminator wall: ICD + CRT both → CRT-D

WHAT IT ENCODES

When a patient meets BOTH ICD criteria and CRT criteria (LVEF $\leq 35\%$, sinus rhythm, NYHA II–IV on GDMT, with LBBB and QRS ≥ 150 ms giving the strongest benefit), implant a combined CRT-D rather than two separate devices. CRT resynchronizes the ventricles to improve symptoms, EF, and remodeling, while the defibrillator component covers sudden death — addressing both pump failure and arrhythmic risk in one device.

BOARD TRAP

WRONG answer: 'place a plain ICD' in a NYHA III, EF 28%, sinus-rhythm patient with LBBB and QRS 160 ms. Discriminator: this patient meets CRT criteria too, so the correct device is CRT-D; a defibrillator alone misses the symptom/remodeling benefit of resynchronization.
